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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First / Second Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21MA201 / 20MA201 - ENGINEERING MATHEMATICS - II

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$.	6	L*1	1	1
b)	Obtain the orthogonal trajectories of the family of curves $r^n = a \sin n\theta$.	7	L2	1	1
c)	Using the method of variation of parameters solve the differential equation $(D^2 - 4D + 4)y = \frac{e^{2x}}{x}$.	7	L3	2	2
2. a)	Solve the differential equation $(D^2 - D - 2)y = 36xe^{2x}$.	6	L1	2	1
b)	Solve the differential equation $(4D^2 - 1)y = e^{\frac{x}{2}} + 12e^x + 4$.	7	L2	2	1
c)	A body originally at 80°C cools down to 60°C in 20 minutes. The temperature of the air being 40°C . Find the temperature of the body after 40 minutes from the original.	7	L3	1	2
3. a)	Find the singular and general solutions of the differential equation $y = xp + \frac{a}{p}$.	6	L1	1	1
b)	Solve $x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} + 2y = 10\left(x + \frac{1}{x}\right)$.	7	L3	2	2
c)	Solve $x^2 p^2 + xyp - 6y^2 = 0$.	7	L2	1	1
Unit – II					
4. a)	Find $L\{e^{-3t} \sin 5t \sin 3t\}$.	6	L1	3	1
b)	If $f(t)$ is a periodic function with period T , then prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$.	7	L2	3	1
c)	Solve $L^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\}$.	7	L3	4	2
5. a)	Find $L^{-1}\left\{\frac{5e^{-3s}}{s} - \frac{e^{-s}}{s}\right\}$.	6	L1	4	1
b)	State and prove the convolution theorem of Laplace inverse transforms.	7	L2	4	1
c)	Determine $L\{t^2 e^{-3t} \sin 2t\}$.	7	L3	3	2
6. a)	Find $L\left\{\int_0^t \frac{1-\cos 2t}{t} dt\right\}$.	6	L1	3	1
b)	Solve $x''(t) + 4x'(t) + 4x(t) = 4e^{-2t}$, given $x(0) = -1$, $x'(0) = 4$.	7	L2	4	1
c)	Rewrite the following using unit step function and find its Laplace transforms. $f(t) = \begin{cases} t-1, & 0 \leq t < 2 \\ 3-t, & 2 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$	7	L3	3	2
Unit – III					
7. a)	Form the partial differential equation by eliminating arbitrary function $z = f(x^2 - y^2)$.	6	L1	5	1
b)	Derive one dimensional Heat equation.	7	L2	5	1

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- c) Solve the following partial differential equation by direct integration.

$$\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x - y) = 0$$

7 L3 5 2

8. a) Solve the following non-linear PDE $yp + xq + pq = 0$.
 b) Derive one dimensional wave equation.
 c) Solve the following linear partial differential equation by Lagrange's method $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$.

6 L1 5 1

7 L2 5 1

7 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
